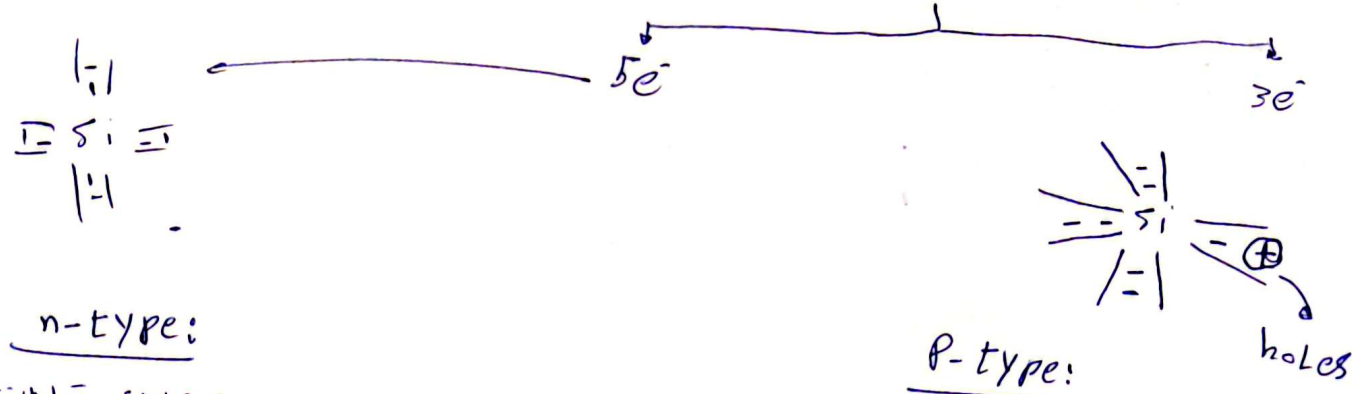


$$R = \frac{\rho L}{A}$$

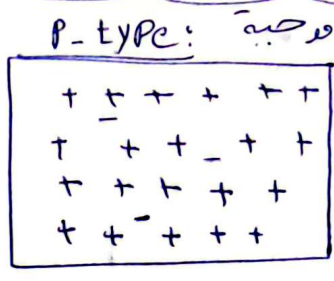
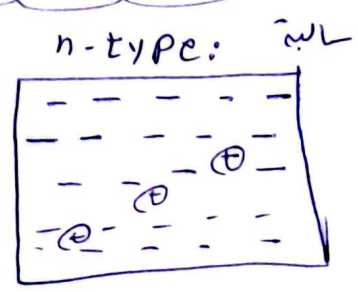
* 1- Intrinsic Material: مواد تم تصنيعها (مواد نقية) (شوائب منها)

* 2- extrinsic material: → مواد تم إضافة شوائب إليها بكميات محدودة



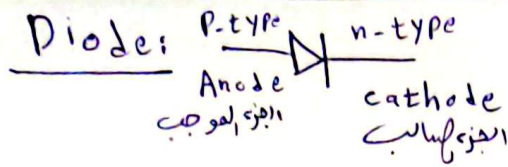
n-type:
عازلة شوائب تم إضافة
(Si, Ge)
(e) لها وزاد
(e)

p-type:
عازلة شوائب تم إضافة
(Si, Ge) لها ونقص (e)



majority: (electron) (-) الغالبية	majority: (holes) (+)
minority: (holes) (+) القلية	minority: (electron) (-)
donor atoms	acceptor atoms

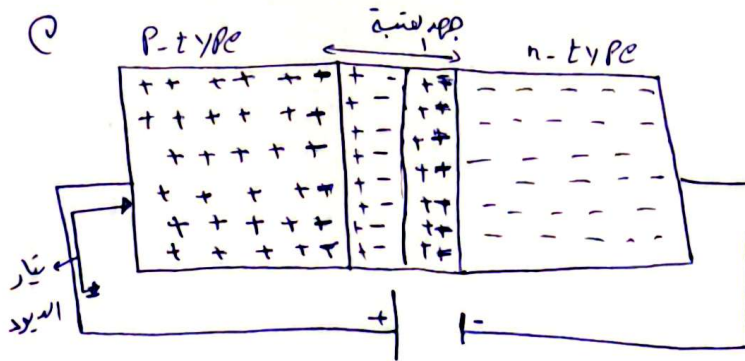
* C.N.F *



طالعة انجيز, Diode

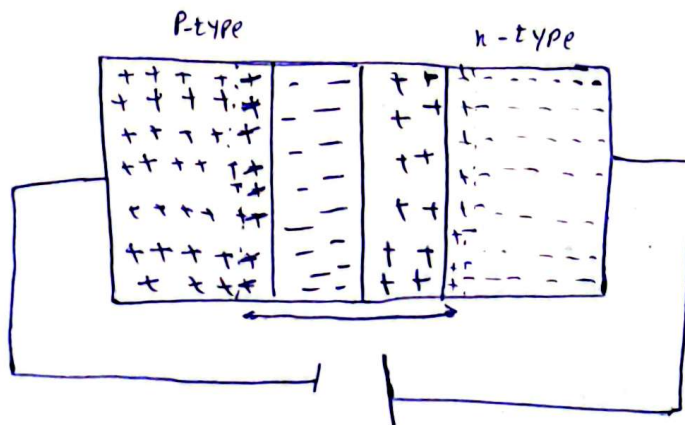
① Forward Biasing:

② reverse Biasing:

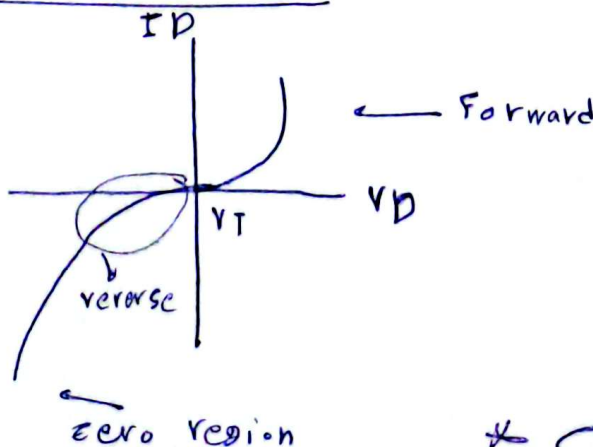


* انما كان جهد البطارية اقل من جهد المنطقة (P) تكون على ارجاء

② reverse Biasing:



* Diode characteristics:

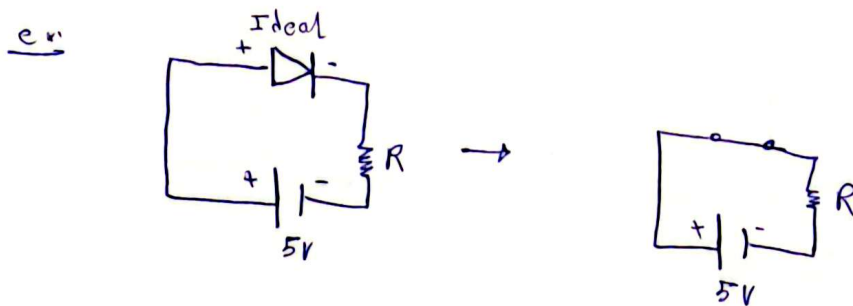
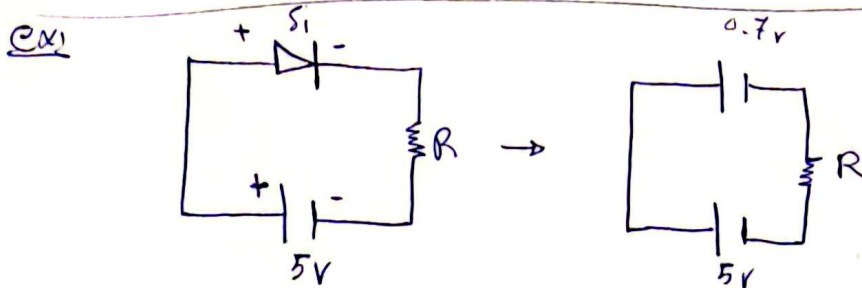


* $C \cdot N \cdot F$ *

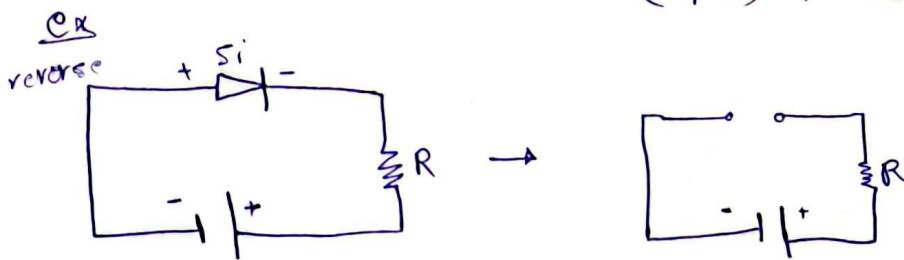
- ③ مثال في اشتقاق Diode 1-8 يكون صرحاً Forward اختيار (مادي)
 2- الجهد العكسي على Diode عبر منحنى الجهد (V-T)

* V_T : @ $S_i = 0.7$
 ② $G_e = 0.3$
 ③ $I_{del} = 0V$

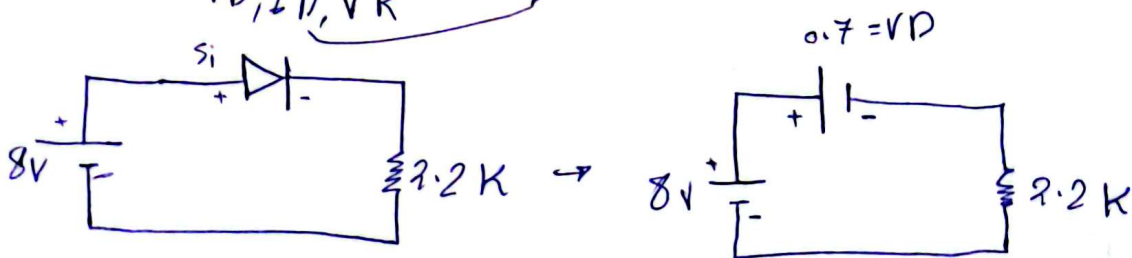
نستبدل Diode ببطارية
 $S_i = 0.7$
 $G_e = 0.3$
 $I_{del} = S_i = 0.7$



إذا افترضنا أن Diode (op.s) →



ex4 Find V_D, I_D, V_R $V_D = I_D R$



$$V_D = 0.7$$

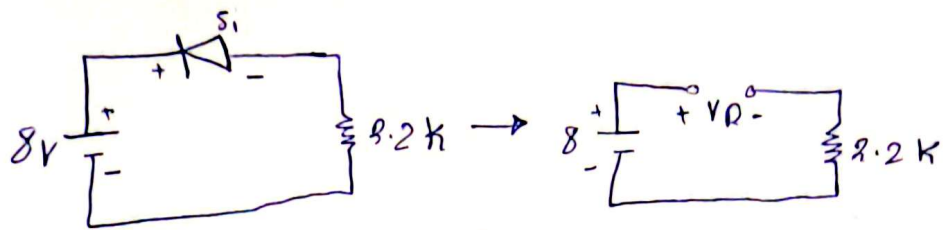
$$I_D = -5 + 0.7 + I(2.2K)$$

$$\frac{2.2K}{2.2K} I = \frac{4.3}{2.2K} = 1.95 \times 10^{-3}$$

$$V_R = I \cdot R = 1.95 \times 10^{-3} \cdot 2.2K = 4.3V$$

C.N.E

ex: Find V_{DQ} , I_{DQ} , V_R



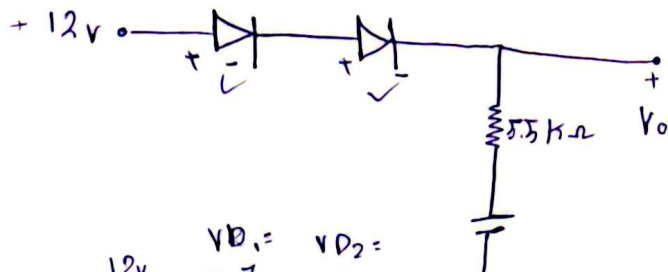
$$I_D = 0$$

$$V_R = I \cdot R = 0$$

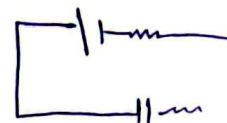
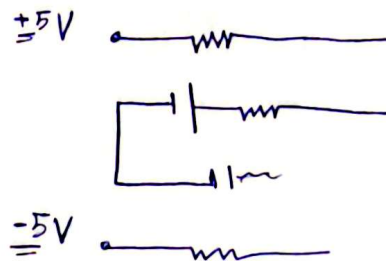
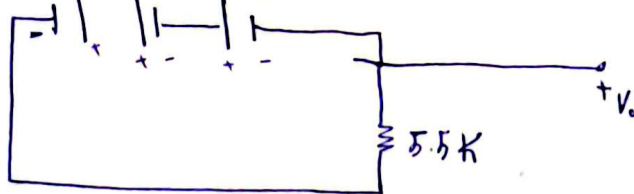
$$V_{DQ} \Rightarrow -8 + V_D + V_R = 0$$

$$V_{DQ} = 8V$$

ex: Find V_D , V_{D2} , I_D , V_R , V_o



$$V_{D1} = 0.7V, V_{D2} = 0.3V$$



$$V_{D1} = 0.7$$

$$V_{D2} = 0.3$$

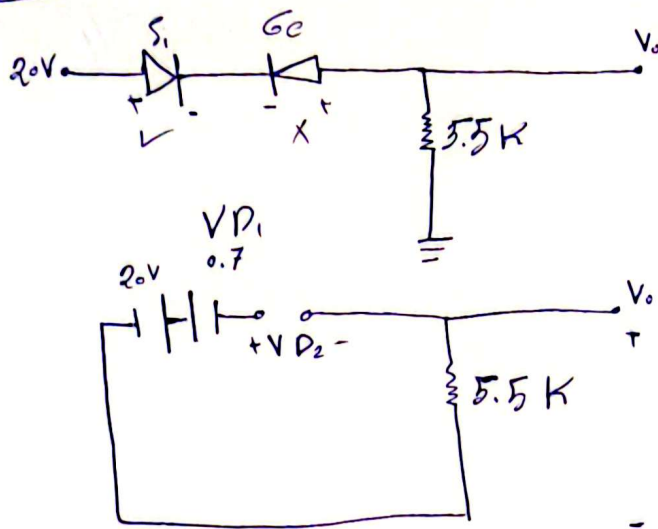
$$I_D = -12 + 0.7 + 0.3 + I(5.5k)$$

$$5.5 \times 10^3 I = 11 \Rightarrow I = 2 \times 10^{-3}$$

$$V_R = I \cdot R = 11V$$

A.C.N.E.A

Ex.



Find $I_D, V_{D1}, V_{D2}, V_R, V_0$

$$V_{D1} = 0.7$$

$$I_D = 0 \text{ (op.s)}$$

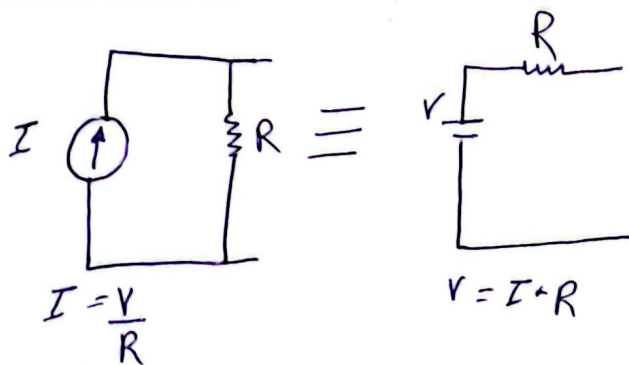
$$V_R = I \cdot R = 0$$

$$V_0 = \text{zero}$$

$$V_{D2} = -20 + 0.7 + V_{D2} + (0 \cdot 5.5)$$

$$\boxed{V_{D2} = 19.3V}$$

* Source Transformation

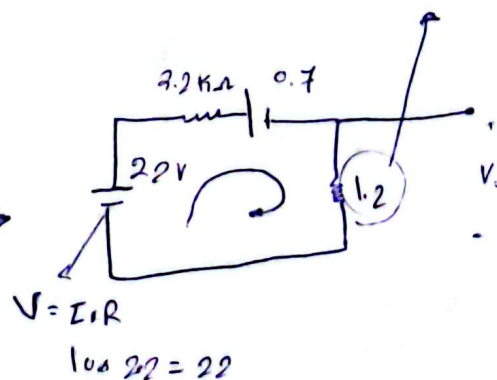
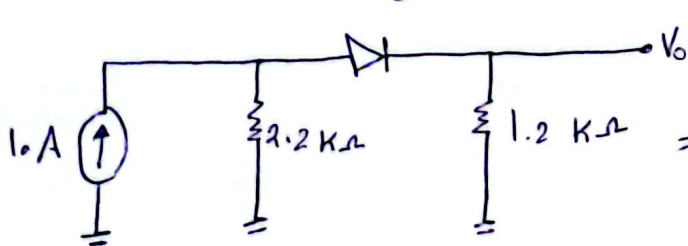


$$V_R = V_0 = I \cdot R$$

$$= 6.26 \text{ mA} \cdot 1.2 \text{ K}$$

$$\boxed{= 7.512 \text{ V}}$$

Ex Find V_0 and I_D



$$-22 + 2.2 \text{ K}(I) + 0.7 + 1.2 \text{ K}(I) = 0$$

$$\frac{(3.4 \text{ K})}{3.4 \text{ K}} I = \frac{21.3}{3.4 \text{ K}} = I = 6.26 \text{ mA}$$

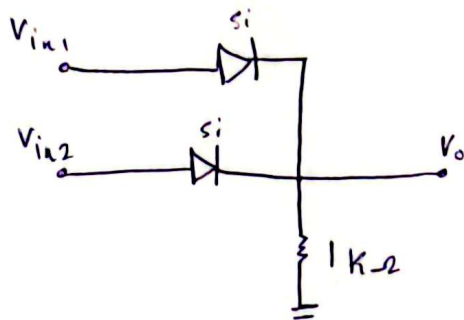
* C.N.F *

* And - OR Gate *

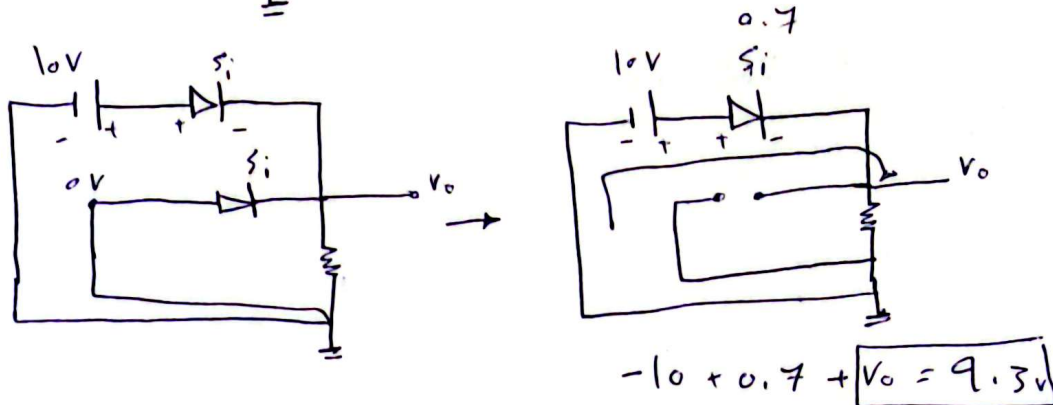
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	1

A	B	$A \text{ and } B$
0	0	0
0	1	0
1	0	0
1	1	1

Ex: Find D_1, D_2, V_o



V_{in1}	V_{in2}	D_1	D_2	V_o
10	0	✓	✗	9.3 V
0	10	✗	✓	9.3 V
10	10	✓	✓	9.3 V
0	0	✗	✗	0 V



* Rectification: $Ac \rightarrow Dc$

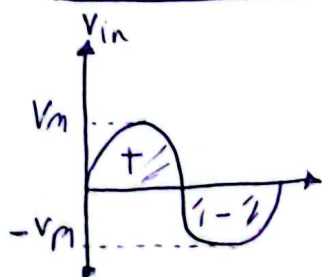
half wave

Full wave

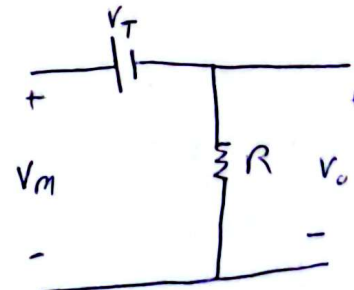
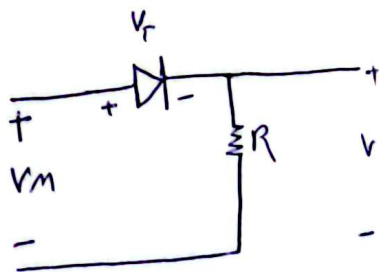
① Bridge

② center tapped

① Half wave:



① +ve half

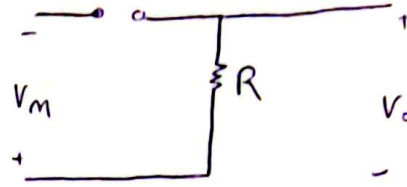
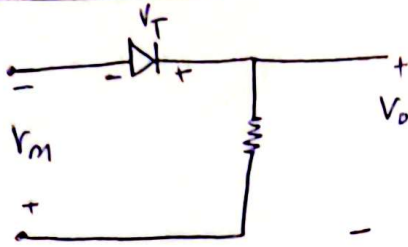


$$-V_m + V_T + V_o = 0$$

$$V_o = V_m - V_T$$

A.C.N.E.K

⇒ -ve half:



$$V_o = V_R$$

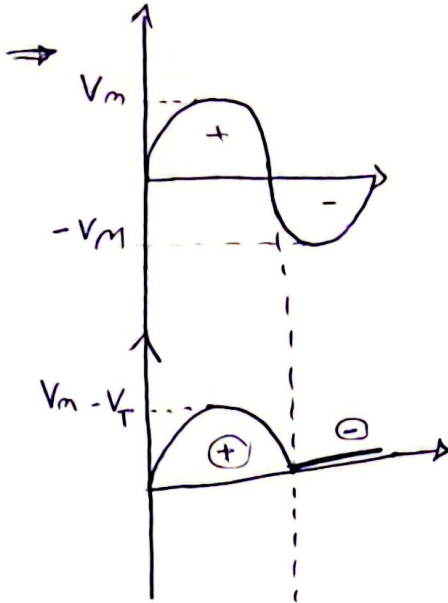
$$= I \times R = 0 \times R$$

$$\boxed{0 = V_o}$$

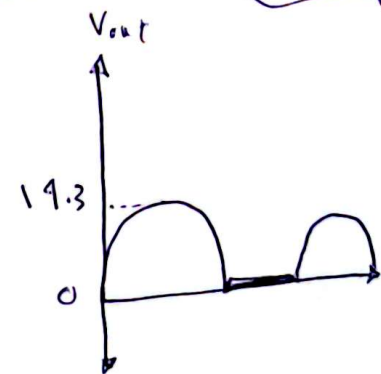
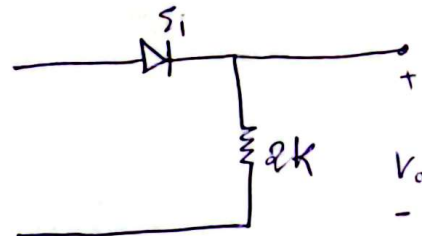
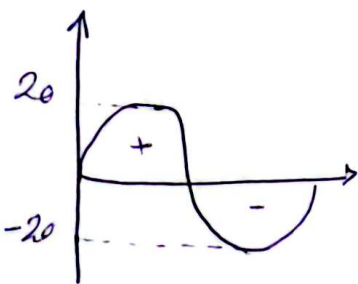
$$* V_{dc} = V_{ava} = \frac{1}{\pi} \times V_o$$

$$= \frac{1}{\pi} \times (V_m - V_T)$$

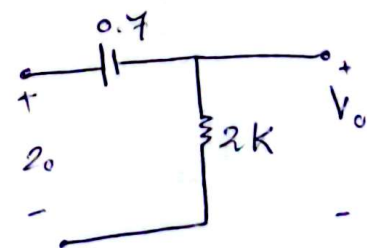
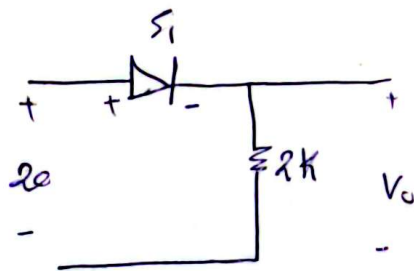
$$* \boxed{f_{in} = f_{out}}$$



ex Find and Draw output signal:



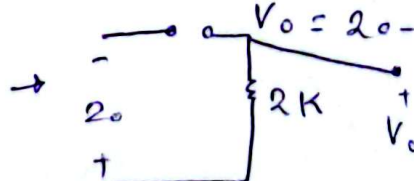
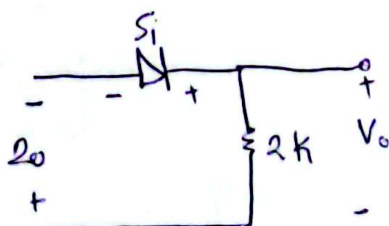
⇒ +ve →



$$-20 + 0.7 + V_o = 0$$

$$V_o = 20 - 0.7 = \boxed{19.3 \text{ V}} \quad V_{o1}$$

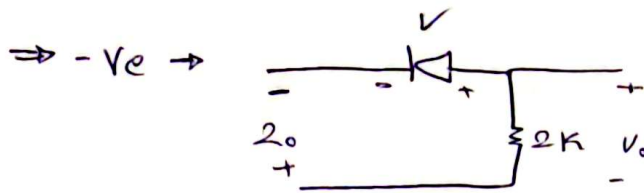
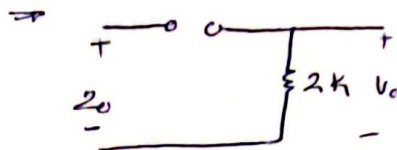
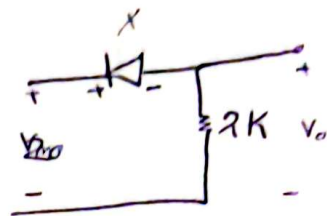
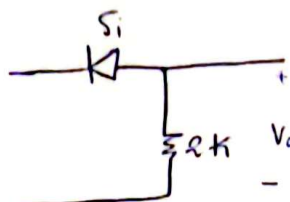
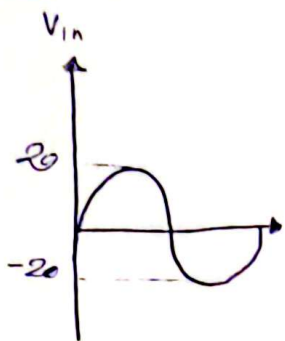
⇒ -ve →



$$V_o = V_R = I \times R = 0 \times R = \boxed{0} \quad V_{o2}$$

* C.N.E *

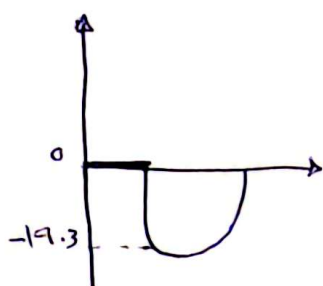
Ex:



$$V_o = V_R = I \times R = 0$$

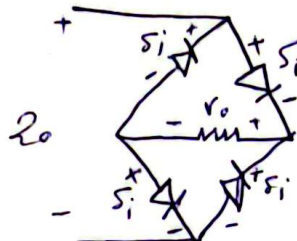
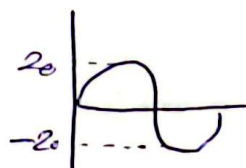
$$+20 - 0.7 + V_o = 0$$

$$V_o = -19.3$$



② Full wave:

① Bridge



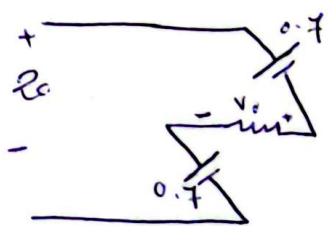
دکتر (2) به شکل و به او بگو

مقابلیت

$$V_{dc} / V_{avg} = \frac{2}{\pi} \times V_o$$

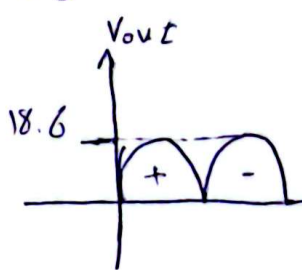
$$f_{out} = 2 f_{in}$$

→ +ve

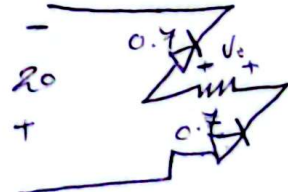


$$-20 + 0.7 + V_o + 0.7 = 0$$

$$V_o = 18.6 \text{ V}$$



→ -ve



$$+20 - 0.7 - V_o - 0.7 = 0$$

$$V_o = 18.6 \text{ V}$$

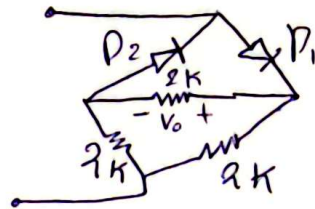
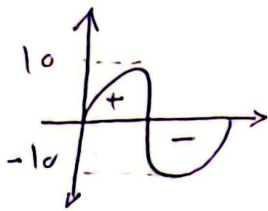
② C.N.E

* Voltage divider Rule:

$$\therefore V_{R_1} = \frac{V \times R_1}{R_1 + R_2}$$

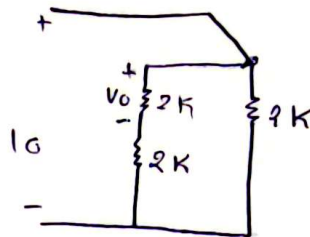
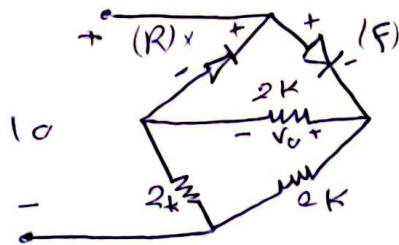
$$\therefore V_{R_2} = \frac{V \times R_2}{R_1 + R_2}$$

ex:



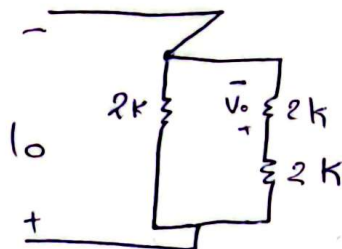
(D_1, D_2 is Ideal)
= $F \rightarrow sh.c$

$\Rightarrow +V_e \rightarrow$

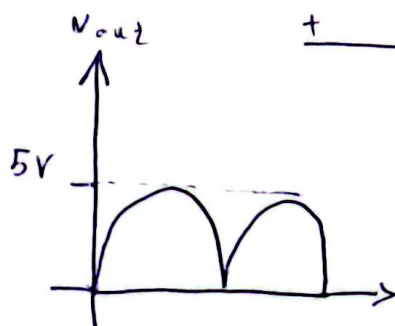


$$V_o = \frac{10 \times 2}{2 + 2} = \boxed{5V}$$

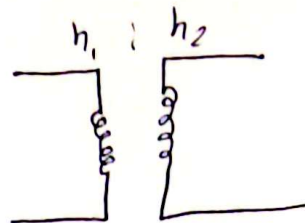
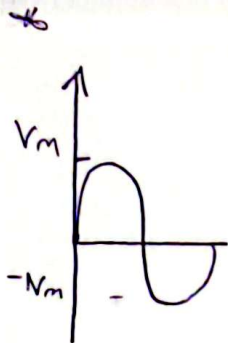
$\Rightarrow -V_e \rightarrow$



$$V_o = \frac{10 \times 2}{2 + 2} = \boxed{5V}$$

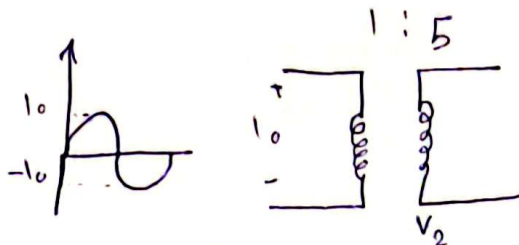


* C.N.E *



$$\frac{V_1}{V_2} = \frac{h_1}{h_2}$$

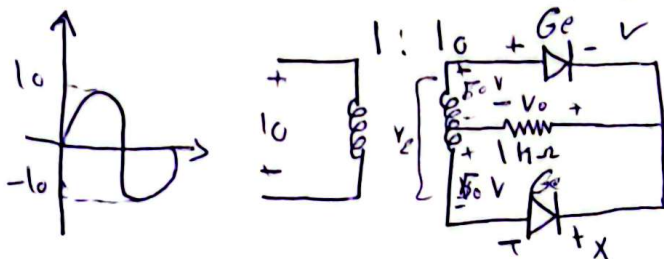
ex1



$$\frac{10}{V_2} = \frac{1}{5} \Rightarrow V_2 = 50V$$

* **B** center tapped transformer:

ex1 Find and Draw the output signal



$$V_{dc} = \frac{2}{\pi} V_o$$

$$f_{out} = 2 f_{in}$$

$\Rightarrow +V_e \rightarrow$

$$\frac{V_1}{V_2} = \frac{h_1}{h_2} = \frac{10}{V_2} = \frac{1}{10} \Rightarrow V_2 = 100V$$

$$\frac{V_o}{2} = 50V$$

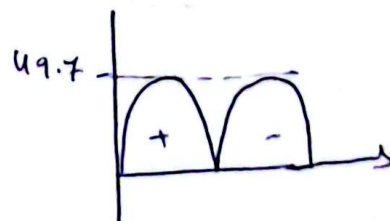
$$\rightarrow \begin{array}{|c|} \hline 0.3V \\ \hline \end{array} \begin{array}{|c|} \hline +50V \\ \hline \end{array} \begin{array}{|c|} \hline -V_o \\ \hline \end{array} \rightarrow -50 + 0.3 + V_o = 0$$

$$V_o = 49.7V$$

$\Rightarrow -V_e \rightarrow$

$$\begin{array}{|c|} \hline -V_o \\ \hline \end{array} \begin{array}{|c|} \hline +50V \\ \hline \end{array} \begin{array}{|c|} \hline 0.3V \\ \hline \end{array} \rightarrow -50 + 0.3 + V_o = 0$$

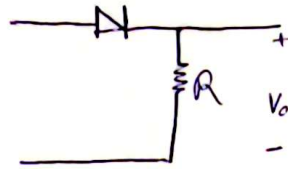
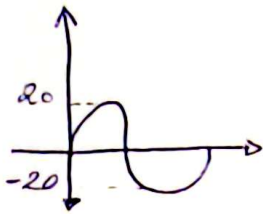
$$V_o = 49.7V$$



* C.N.E *

* Filters: صيغة: العلاقة بين V_p و V_r و V_{DC}

المسوحة ضوئياً بـ CamScanner



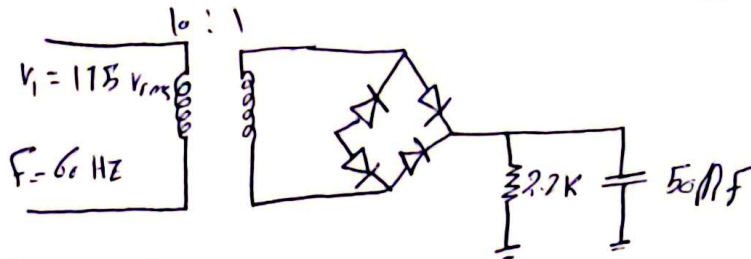
$$V_r(P-P) = \frac{V_p(peak)}{F_o \times R_L \times C}$$

$$V_{DC} = V_p(peak) \left(1 - \frac{1}{F_o \times R_L \times C} \right)$$

$$V_{DC} = V_p(peak) - \frac{V_r}{2}$$

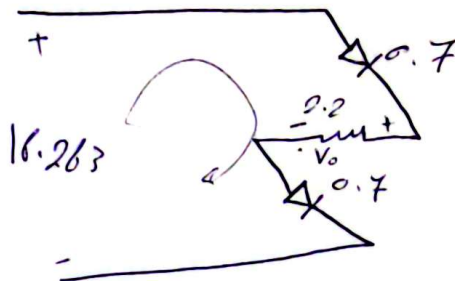
$$r = \frac{V_r(P-P)}{V_{DC}}$$

ex: Find ripple factors for the circuit (ALL diode is Si):



$$V_{rms} = \frac{V_m}{\sqrt{2}} = 115 \times \sqrt{2} = 162.6 = V_m$$

$$\frac{V_1}{V_2} = \frac{n_1}{n_2} = \frac{162}{V_2} = \frac{10}{1} \Rightarrow V_2 = 16.263V$$



$$-16.263 + 0.7 + V_o + 0.7 = 0$$

$$V_o = 14.863$$

$$V_r(P-P) = 14.863$$

$$F_o = 2 F_{out} = 2 \times 60 = 120 \text{ Hz}$$

$$V_r(P-P) = \frac{14.863}{120 \times 2.2 \times 10^3 \times 50 \times 10^{-6}} = 1.1259$$

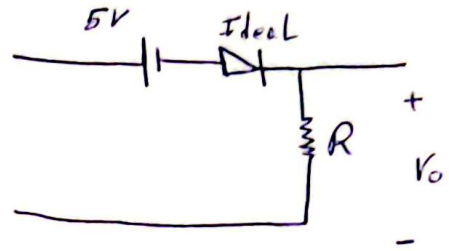
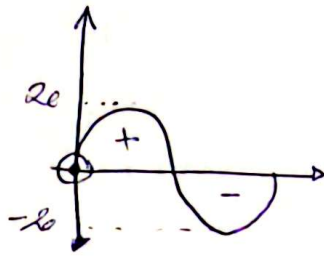
$$* C = N \cdot E *$$

المسألة

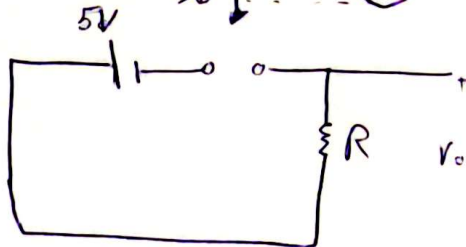
Q5.c → V_o Find and Draw the output signal

② $+V_e \Rightarrow V_o$

③ $-V_e \Rightarrow V_o$

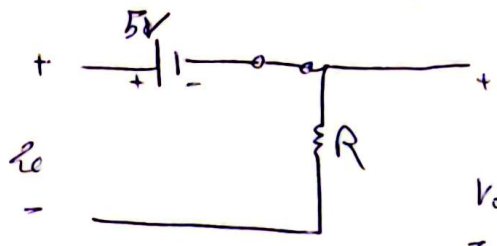


① S.c →



$$V_o = V_R = I_R R = \boxed{0}$$

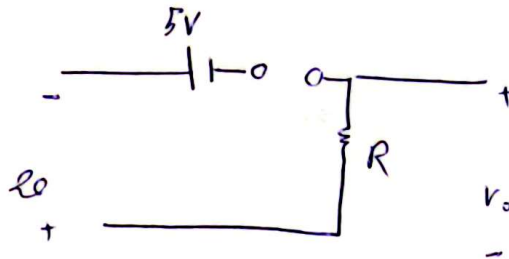
② $+V_e \rightarrow$



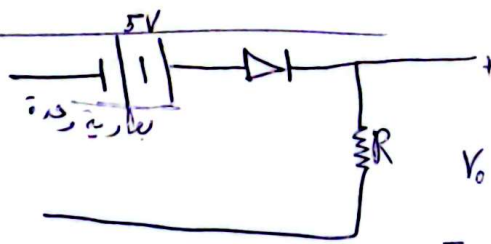
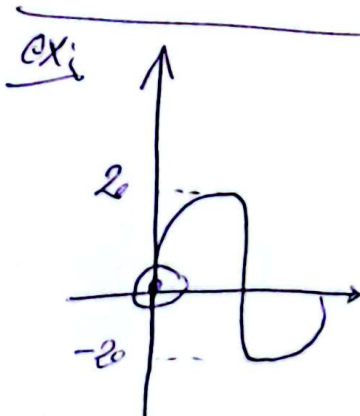
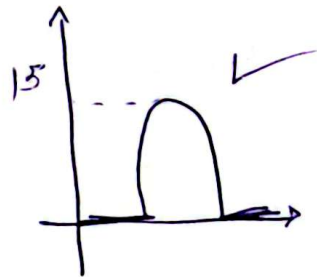
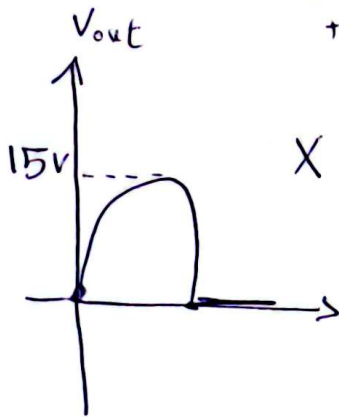
$$V_e - 20 + 5 + V_o = 0$$

$$V_o = \boxed{15V}$$

③ $-V_e \rightarrow$

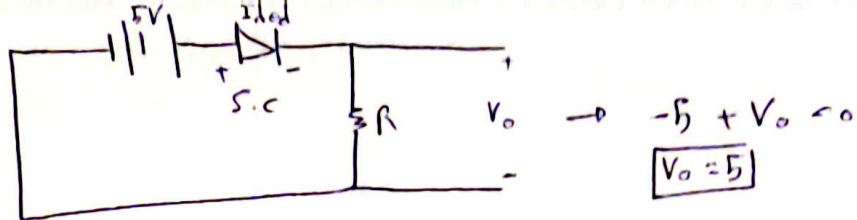


$$V_o = V_R = I_R R = \boxed{0}$$

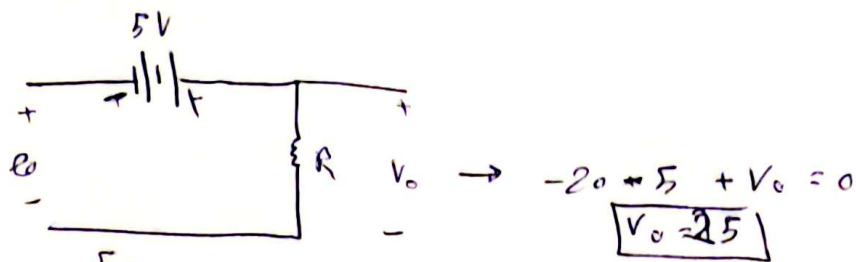


المسألة

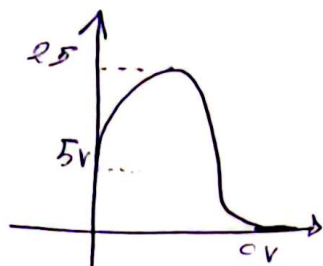
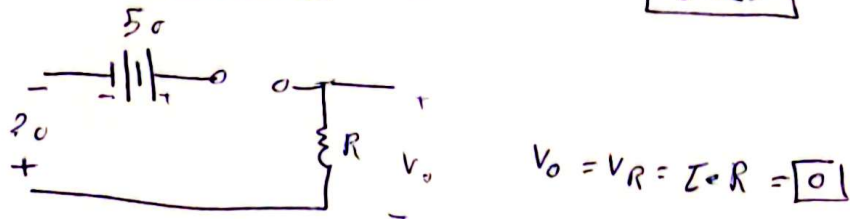
① S.C →



② +V_e →



③ -V_e →



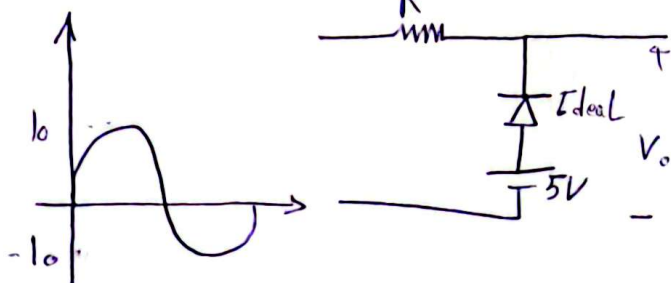
نظارتی دیوید (میکروپروسیسر)

V_o

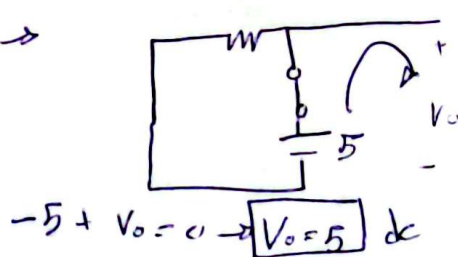
① S.C → V_o

$$d_c = \frac{1}{2} A$$

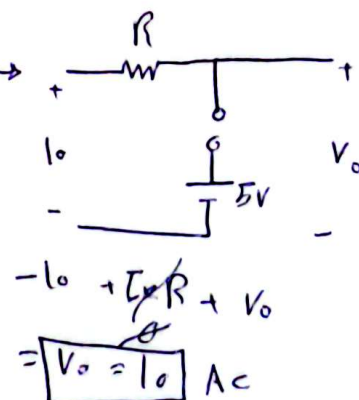
کیرش



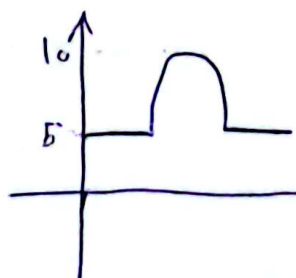
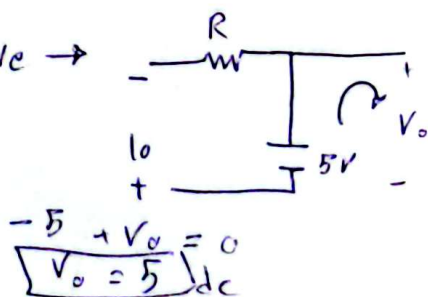
① S.C →



② +V_e →

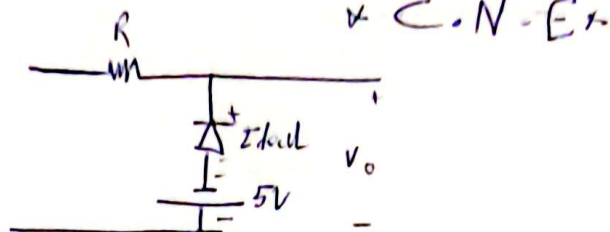
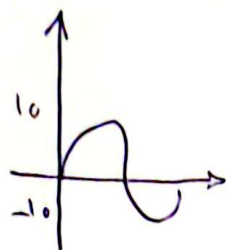


③ -V_e →

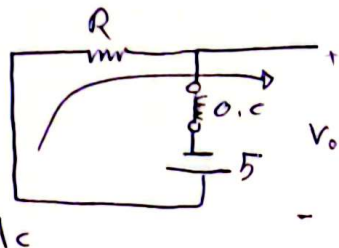


C.N.E

ex_i

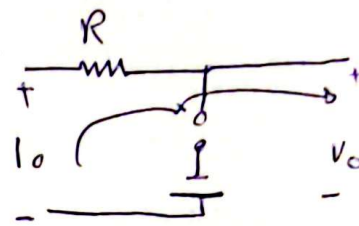


① S.C →



$$V_o = 0 \text{ A.C}$$

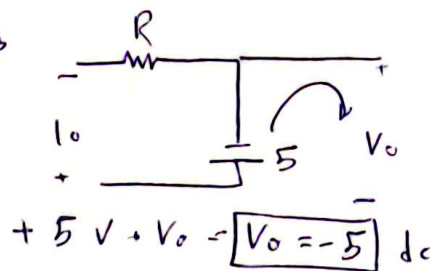
② +ve →



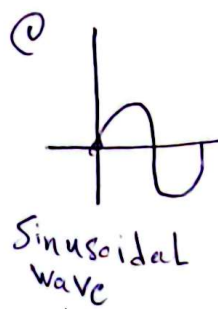
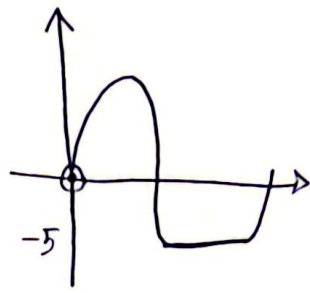
$$-10 + I \cdot R + V_o = 0$$

$$V_o = 10 \text{ A.C}$$

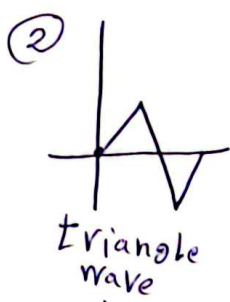
③ -ve →



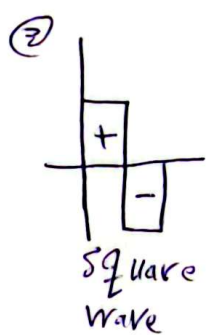
$$+5 \text{ V} + V_o = 0 \Rightarrow V_o = -5 \text{ dc}$$



Sinusoidal wave



Triangle wave

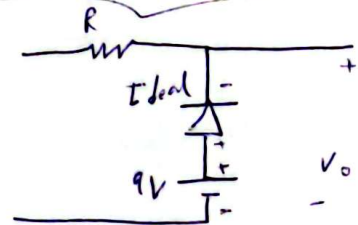
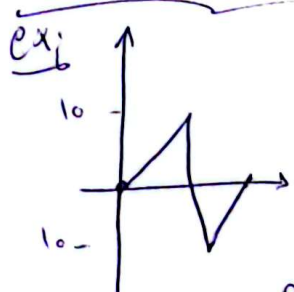


Square wave

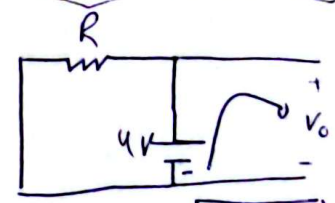
① +ve
② -ve

في موجبة out تؤخذ نفس شكل موجبة in

① S.C
② +ve
③ -ve

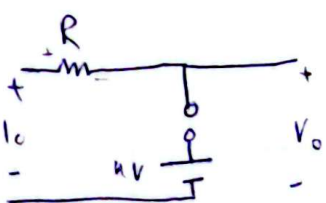


① S.C →



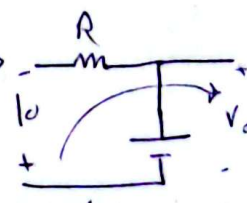
$$-9 + V_o = 0 \Rightarrow V_o = 9 \text{ dc}$$

② +ve →

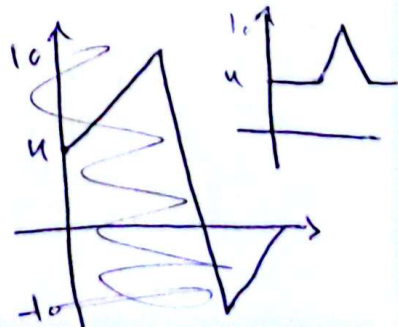


$$-10 + I \cdot R + V_o = 0 \Rightarrow V_o = 10$$

③ -ve →



$$+10 + I \cdot R + V_o = 0 \Rightarrow V_o = -14$$



* Clamper :

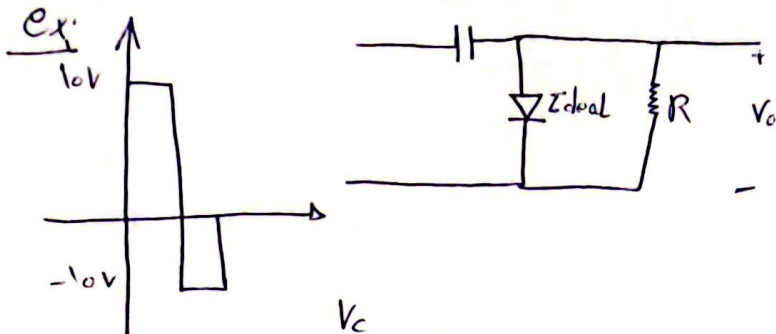
نفسها ما تتغير (V_{p-p}) نفسها ما تتغير

① (f) Diode جزء $\Rightarrow V_o, V_c$

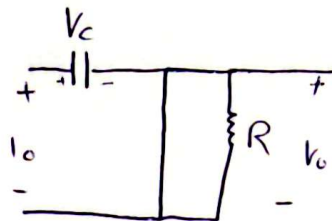
② نذكر الجزء الثاني المتغير

③ لا نذكر $S.C$ وادجه V_o للعبية

Sinusoidal, triangle



① $+V_c \rightarrow$

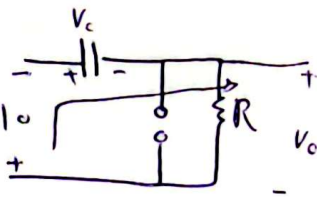


$$V_i = V_R = \text{Zero}$$

$$-10 + V_c = 0$$

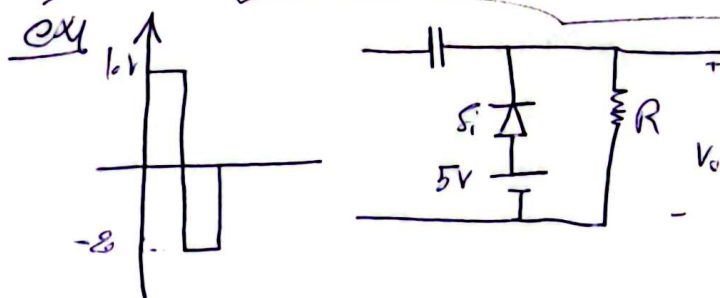
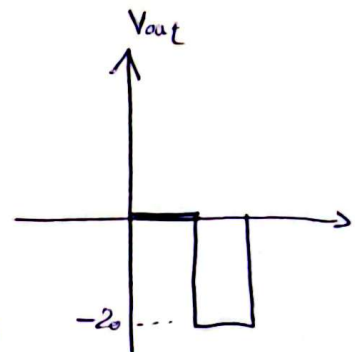
$$V_c = 10$$

② $-V_c \rightarrow$

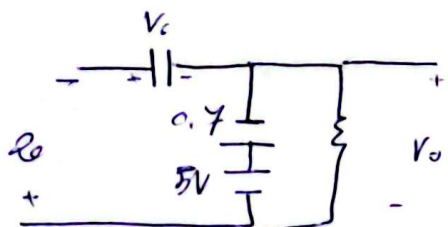


$$+10 + V_c + V_o = 0$$

$$V_o = -20$$



① $-V_c \rightarrow$



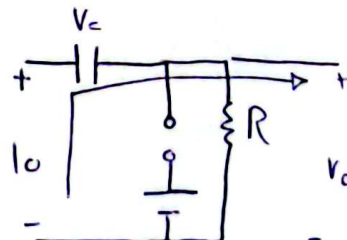
$$+20 + V_c + 0.7 + 5 = 0$$

$$V_c = +24.3$$

$$+20 + V_c + V_o + 0.7 - 5 = 0$$

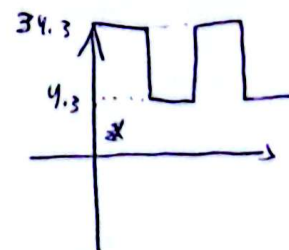
$$V_o = 4.3$$

② $+V_c \rightarrow$



$$-10 + V_c + V_o = 0$$

$$V_o = 34.3$$

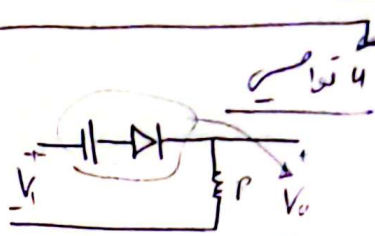
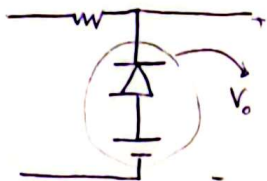


* C.N.F *

Design:

@ Clipper:

الانوار



اذا ابلست الموجة بقية غير صفر
و ثبتت بعد بيت تغيرت

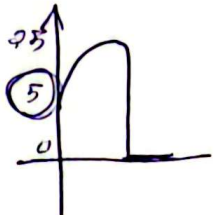
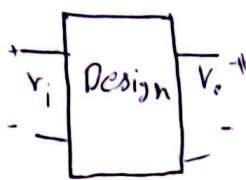
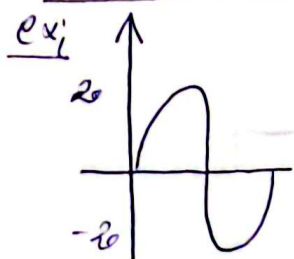
اذا ابلست الموجة out بقية غير صفر وتظل القيمة مباشرة



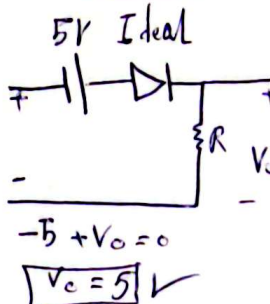
في حال ابلست من صفر بعد بيت ثبتت



اذا الموجة فيها قصي



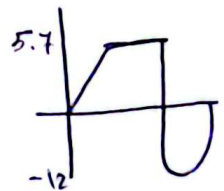
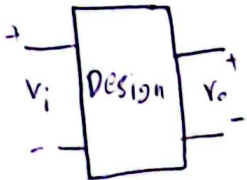
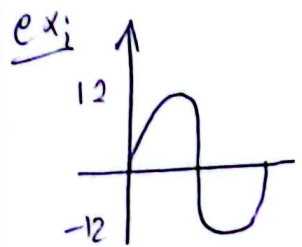
$\Rightarrow SOL \Rightarrow$



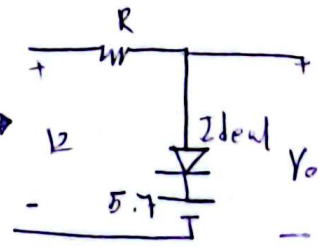
$$-5 + V_o = 0$$

$$V_o = 5V$$

تأثير بطارية
Diode (Reverse) $\Leftarrow$ غير صفر
Diode (Forward) $\Leftarrow$ غير صفر



$\Rightarrow SOL \Rightarrow$



* C.N.E *

② C Lammers: طارية , فولت , ديود

17

إذا موجة output أغلبها بالقيمة السالبة (تحت)



اتجاه للديود (تحت)

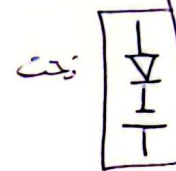
إذا موجة output أغلبها بالقيمة الموجبة (فوق)

اتجاه للديود (فوق)



اتجاه للبطارية

إذا كانت الموجة فوق (تكون اتجاه البطارية موجبة) :-



تحت



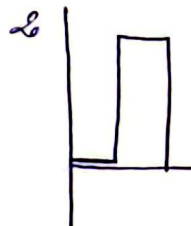
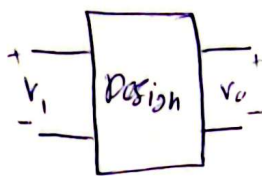
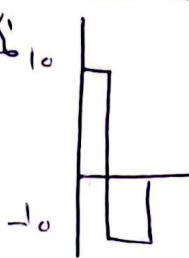
فوق

إذا كانت الموجة على محور السينات لا يوجد بطارية

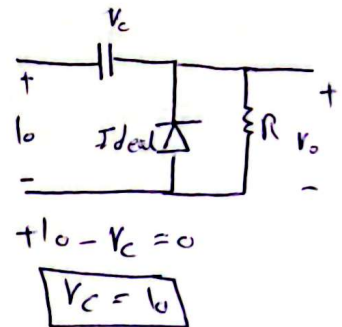
قيمة البطارية هي أقل قيمة من موجة (الادوات)

قيمة الموجة من خلال المحور

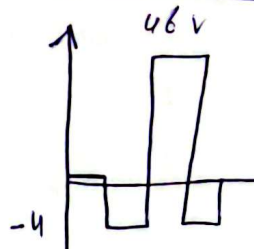
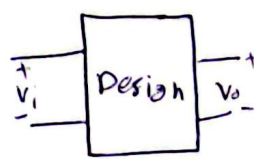
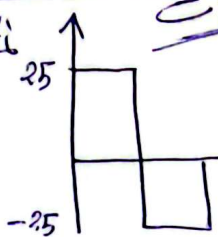
ex 10



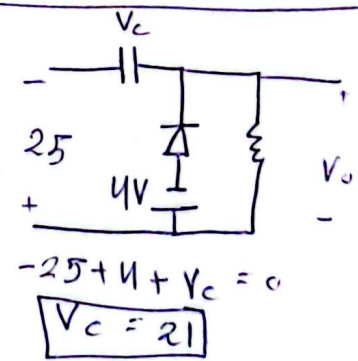
sol



ex 11



sol



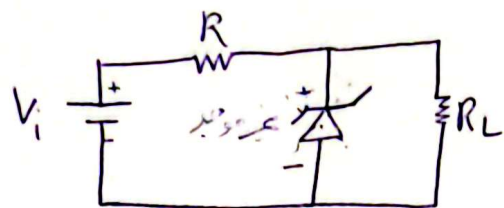
* C.N.E *

* Zener Diode:

① Fixed (V_i, R_L)

$$C \propto N \cdot E$$

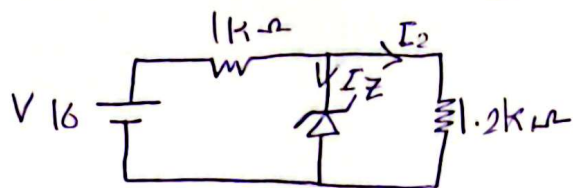
$$V_Z < V \quad \text{في حالة reverse bias}$$



$$\Rightarrow V_{RL} = V_i \times \frac{R_L}{R_L + R} \quad \boxed{V_{RL} = V_Z} \quad \text{if } V_{RL} \geq V_Z$$

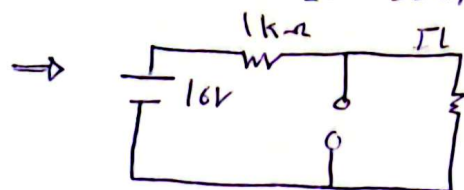
$$\boxed{Z = 0} \quad \text{if } V_{RL} < V_Z$$

ex: Find V_R, V_{RL}, I_Z, P_Z



$$\Rightarrow \text{so } V_{RL} = \frac{16 \times 1.2 \text{ k}}{1 \text{ k} + 1.2 \text{ k}} = \boxed{8.72 \text{ V}}$$

$$V_{RL} < V_Z$$

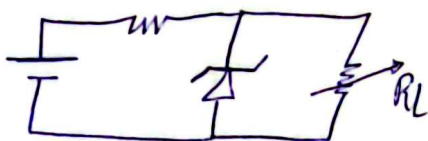


$$\Rightarrow V_R = \frac{16 \times 1}{1 + 1.2} = \boxed{7.273 \text{ V}}$$

$$\boxed{I_Z = 0}$$

$$\boxed{P = I \times V = 0}$$

② Fixed (V_i, R_L)



Find

$$\boxed{R_{L \min}, R_{L \max}}$$

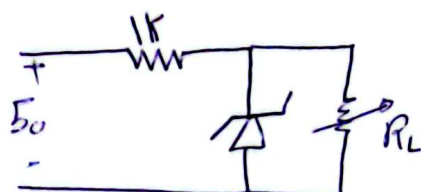
$$* R_{L \min} = \frac{V_Z \times R}{V_i - V_Z}$$

$$- I_R = \frac{V_i - V_Z}{R}$$

$$- I_{L \min} = I_R - I_{ZM}$$

$$- R_{L \max} = \frac{V_Z}{I_{L \min}}$$

ex:



$$I_{ZM} = 32 \text{ mA}$$

$$V_Z = 10 \text{ V}$$

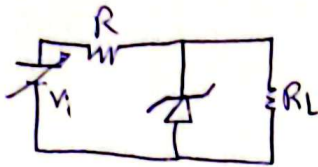
$$R_{L \min} = \frac{V_Z \times R}{V_i - V_Z} = \frac{10 \times 1 \text{ k}}{50 - 10} = \boxed{250 \Omega}$$

$$I_R = \frac{V_i - V_Z}{R} = \frac{50 - 10}{1 \text{ k}} = \boxed{40 \text{ mA}}$$

$$I_{L \min} = I_R - I_{ZM} = 40 \text{ mA} - 32 \text{ mA} = \boxed{8 \text{ mA}}$$

$$R_{L \max} = \frac{V_Z}{I_{L \min}} = \frac{10}{8 \text{ mA}} = \boxed{1250 \Omega}$$

3] V_i منسوب، R_L ثابت

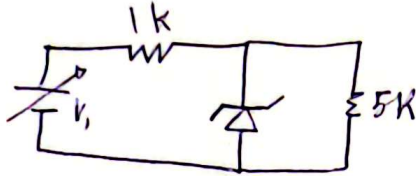


$$* V_{in \min} = \frac{V_Z(R + R_L)}{R_L}$$

$$* I_{R \max} = I_{ZM} + I_{RL}$$

$$* V_{in \max} = (I_{R \max} * R) + V_Z$$

ex1



$$V_Z = 10V$$

$$P_{ZM} = 30mW$$

$$- V_{in \min} = \frac{10 * (1K + 5K)}{5K} = 12V$$

$$- I_{R \max} = I_{ZM} + I_{RL} =$$

$$P_{ZM} = V_Z * I_Z$$

$$I_Z = \frac{P_{ZM}}{V_Z} = \frac{30m}{10} = 3m$$

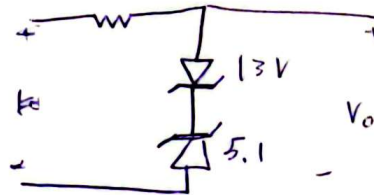
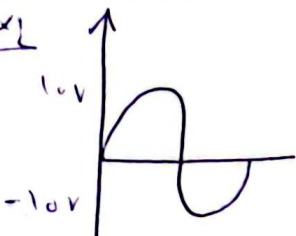
$$V_{RL} = I_{RL} * R_L$$

$$I_{RL} = \frac{V_{RL}}{R_L} = \frac{10}{5K} = 2m$$

$$- I_{R \max} = 2m + 3m = 5m$$

$$- V_{in \max} = (5m * 1K) + 10 = 15V$$

ex2

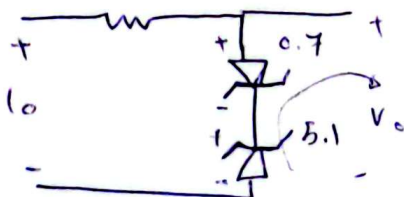


① I_Z at Zener Voltage at forward is (0.7V)

Find ① V_o at +ve half

② V_o at -ve

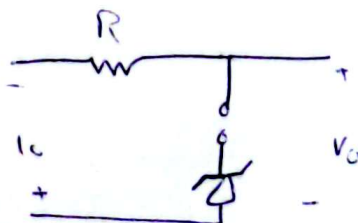
① Sol $\Rightarrow +ve$



$$-5.1 - 0.7 + V_o = 0$$

$$V_o = 5.8V$$

② Sol $\Rightarrow -ve$



$$+10 + \frac{V_o}{R} + V_o = 0 \Rightarrow V_o = -10$$

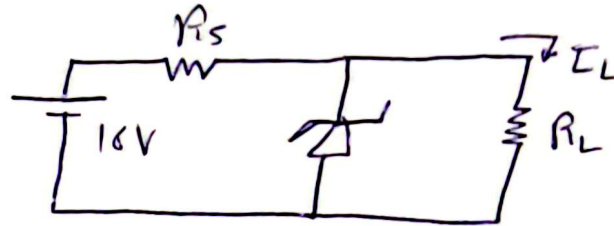
* Design 1

ex) Design the network to maintain V_L at 12V for load variation I_L

From (0 mA) to (200 mA) that determine R_S and V_Z

sol. $V_L = 12 = V_Z$

$I_L = 200 \text{ mA}$



$$V_{RL} = I_L \times R_L$$
$$12 = 200 \text{ mA} \times R_L$$

$$R_L = \frac{12}{200 \text{ mA}} = 60 \Omega$$

$$V_{RL} = \frac{V_i \times R_L}{R_L + R_S}$$
$$12 = \frac{16 \times 60}{60 + R_S}$$

$$R_S = 20 \Omega$$

$$* C = N \times E *$$